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Research Statement

Research interests: Applied Microeconomics, Production Economics, and Industrial Organization.

I am an applied microeconomist studying productivity and technology adoption, with expertise in production econometrics. My research asks why technology investments generate uneven productivity gains and whether improved firm performance reflects greater efficiency or greater pricing power. These questions matter for policies that support digital adoption and smaller firms, and for understanding financial intermediation. I develop methods that account for firms' input decisions and apply them to manufacturing and banking. My agenda connects these economic questions with research on causal mechanisms and reliable productivity measurement.

Job Market Paper: Digitalization, Factor-Biased Productivity, and Market Power

Higher revenues do not reveal how digital investments change production. Technology may help workers produce more with the same hours, improve efficiency across inputs, or enable firms to charge higher prices relative to costs. Distinguishing these channels helps explain what adoption achieves and how to evaluate policies that subsidize it. My job market paper, **“Digitalization, Factor-Biased Productivity, and Market Power”** (with Subal C. Kumbhakar), examines these channels within a common framework using Spanish manufacturing firms over 2000–2019.

The main finding is that digitalization is associated with modest improvements in productive efficiency alongside a much smaller change in markups. In the model allowing firms to have market power, a 0.10-point increase in the digitalization index, which ranges from zero to one, is associated with approximately **0.55 percent higher labor-augmenting productivity, 0.43 percent higher Hicks-neutral productivity, and 0.07 percent higher markups** in the following period. The first productivity measure captures the efficiency of labor hours; the second captures efficiency across the input bundle. In the price-taking benchmark, the corresponding productivity estimates are 0.37 and 0.62 percent. These estimates describe conditional relationships under the model's assumptions.

The gains also differ across firms. The labor-efficiency association is positive and statistically significant for small firms under both market environments, and significantly larger than for large firms, whose estimate is imprecise. Among small firms, IT technologies, digital human capital, and automation each have positive labor-efficiency associations in the model with market power. The markup response is significantly smaller in industries with high digital intensity. These findings motivate evaluating adoption programs by the productivity channels, technologies, and firms involved.

I use Spanish manufacturing dataset, Encuesta Sobre Estrategias Empresariales (ESEE). It is rich in production datasets, and digitalization variables. My digitalization index comprises a thirteen-variable index covering IT, digital human capital, automation, and interaction with customers and suppliers. I develop a sequential identification approach that distinguishes labor efficiency, broader productivity, and markup dynamics under imperfect competition. Technology parameters are identified before markups are recovered from input expenditure shares, without requiring firm-level input-price data. This sequence addresses the difficulty that the same shares reflect both technology and pricing behavior.

Productivity and Technological Change in Banking

My publication on banking examines a related puzzle: why can technological progress and consolidation coexist with declining productivity? In **“The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking”** (with Subal C. Kumbhakar; *Economic Modelling*, 2026), I study seventeen Nigerian commercial banks over 2000–2023. The analysis accounts for banks’ multiple outputs and endogenous input choices, distinguishing changes in production possibilities from changes in realized productivity.

I estimate an average annual productivity decline of **3.5 percent**, with steeper declines among larger and internationally licensed banks. Yet technical change reduces costs, and estimated returns to scale average 1.46, indicating substantial potential gains from expansion. Lending is the principal source of cost pressure. The economic lesson is that modernization and greater scale need not be sufficient to improve performance. The findings motivate attention to lending efficiency, credit information, collateral enforcement, and bad-loan resolution. This project extends my productivity research to financial intermediation and highlights the importance of understanding how firms use improved technologies.

Causal Mechanisms and Heterogeneous Effects

Understanding the returns to adoption also requires identifying the channels through which output changes. In **“Mediation Analysis in Production Functions with Unobserved Productivity”** (with Subal C. Kumbhakar), I ask how much of digitalization’s relationship with output operates through changes in labor hours. More productive firms may both adopt technology and adjust employment, making it difficult to distinguish a labor-mediated effect from differences in underlying productivity. The paper develops a framework that controls for unobserved productivity in both the production and labor-hours equations. Simulations show that omitting this control from the labor equation can substantially overstate the indirect effect. An application to Spanish manufacturing connects this methodological issue to the economic distinction between producing more through additional labor and producing more efficiently.

My published paper, **“Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects”** (with Ivan Korolev and Nancy Dhameja; *Economics Letters*, 2026), examines how the structure of treatment-effect variation should guide empirical method choice. In simulations with randomized assignment, forests perform best when effects change sharply, while penalized splines generally perform better with smooth heterogeneity and in designs combining continuous, binary, and group-level covariates. The findings provide practical guidance for researchers studying who benefits from an intervention. They also inform my broader interest in matching empirical methods to the economic quantities researchers want to learn.

Solo Project: Regularization in Translog Production Models

In my solo project, **“Regularization in Translog Production Models,”** I study whether methods that improve output prediction also improve the economic conclusions drawn from production models. Flexible models allow the contribution of an input to vary with a firm’s input mix. Their squared and interaction terms can, however, be highly correlated. Consequently, a model may predict output accurately while producing unstable estimates of the contribution of labor or capital. This instability matters because input elasticities underpin conclusions about returns to scale and production-based markups.

I am developing a comparison of Lasso, Ridge, Elastic Net, principal component regression, and partial least squares that evaluates these economic quantities directly. The planned simulations will vary input correlations, sample size, and the underlying production relationships. I will compare the bias and

variability of input elasticities, the accuracy of returns-to-scale estimates, and the frequency of economically implausible estimates, such as negative marginal products. I will also assess how elasticity errors carry through to inferred markups. A central question is whether choosing regularization for predictive accuracy yields different conclusions from choosing it for elasticity recovery.

The project's goal is to provide guidance on when regularization stabilizes economically relevant estimates and when it conceals distortions. I plan to assess its use alongside methods that address endogenous input choices, since regularization alone does not resolve that problem. This project develops an independent strand of my agenda: evaluating econometric methods by their ability to recover the quantities needed for economic interpretation and policy analysis.

Research Beyond the Dissertation

My next empirical project, **“Artificial Intelligence and Spatial Factor-Biased Productivity”** (with Subal C. Kumbhakar), extends the adoption framework to ask who benefits from AI beyond the adopting firm. AI may improve the efficiency of particular inputs within firms, while knowledge flows, worker mobility, and supply-chain relationships transmit gains to others. I plan to compare exposure through geographic proximity and economic linkages, and to assess whether the framework can distinguish spillovers from common shocks. The empirical design will require information on the AI applications firms use and the production activities they affect.

Beyond the dissertation, I plan to connect heterogeneous treatment-effect methods with production models to examine which firms benefit most from adoption and which complementary investments help explain those differences. A priority is to accommodate endogenous adoption and input choices while retaining interpretable productivity measures. My solo project will support this agenda by developing criteria for selecting methods based on the reliability of elasticities and related economic outcomes. Together, these projects will examine how technology improves productive efficiency, why its benefits are uneven, and how gains extend across firms and industries.

References

Publications

- [1] “Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects,” with Ivan Korolev and Nancy Dhameja. *Economics Letters*, 268, 113205, 2026. doi:10.1016/j.econlet.2026.113205.
- [2] “The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking,” with Subal C. Kumbhakar. *Economic Modelling*, 107859, 2026. doi:10.1016/j.econmod.2026.107859. Replication code.

Job Market Paper

- [3] “Digitalization, Factor-Biased Productivity, and Market Power,” with Subal C. Kumbhakar. Job market paper, September 2026.

Working Papers

- [4] “Mediation Analysis in Production Functions with Unobserved Productivity,” with Subal C. Kumbhakar. Under review at *Economics Letters*.

Work in Progress

[5] “Regularization in Translog Production Models.” Solo-authored.

[6] “Artificial Intelligence and Spatial Factor-Biased Productivity,” with Subal C. Kumbhakar.